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Responsible Oil and Gas Assets Decommissioning: Advantages and Benefits

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Abstract

The decommissioning of oil and gas wells at the end of their economic life is a critical operation in the oil and gas industry. Decommissioning sees an increased scrutiny regarding its long-term environmental impact. It requires full isolation across the entire wellbore including the formation to casing and across all annuli or tubing. To improve time- and cost-efficient abandonment and to provide ‘rock-to-rock’ isolation, the decommissioning requires data and expertise across a large number of areas.

The decommissioning process starts with assessing the wells to understand the asset current conditions and potential risks. The planning for abandonment starts with the analysis of the last operating years operations and well productivity. Followed by a detailed log evaluation to identify existing annular isolation, lack of isolation and potential casing corrosion. Based on this data, an abandonment plan is created to fulfill any internal and regulatory requirements. The plan includes cement plugs to provide isolation across potential hydrocarbon zones and aquifers; casing retrieval / casing cutting; placing appropriate well barriers; and ends with a location restoration and monitoring.

The primary objective is to create a seal that will prevent fluids from migrating past the reservoir sections based on the well depth and geological conditions. To improve efficiency, a detailed plan is prepared to set the cement plugs and required remediations using most of the time specialty tools like perforate, wash and cement in stretches showing poor isolation behind the production casing, casing patches, cement squeezes, down hole tools, among others. Plug lengths and plug placement are always optimized to limit the wait on cement but at the same time to identify, address any operational risks and comply with any necessary isolation testing.

Cement plugs are placed in a method to reduce and eliminate the need for costly aqueous fluids left in the wellbore after the well decommissioning which includes corrosion inhibitors, oxygen scavengers, stabilizers for different depths and temperatures. Proprietary software is utilized to minimize fluid contamination and simulation results are used to optimize lab testing. A detailed validation of well barriers is agreed to comply with regulations to provide long term isolation after the well abandonment and avoid leakage of formation fluids and gases to the atmosphere or surface.

This new process has now been standardized by various operators in the region. As wells are safely and responsibly plugged and abandoned, leaving no concerns regarding the long-term environmental impact

of retired assets, particularly those close to social centers and with any risk of unplanned formation fluid emissions. The novel and optimized approach has provided both a time saving of 36% time and by optimizing the fluid requirements and logistics, the total well abandonment costs has reduced by 28%.

Introduction

The life of an oil or gas well consists out of several stages as described by Barclay (2001). After discovery of oil or gas, a well is drilled to reach the reservoir. This is followed by a wellbore completion before the well is placed into production. Enhanced recovery techniques, such as recompletion or re-entry, can extend the productive life of the well. However, the final step of the well is abandonment and wellbore decommissioning, which includes the wellbore as well as restoring surface facilities. Wellbore abandonment is designed to be permanent, unlike the relatively short producing timespan of an oil or gas well.

Wellbore abandonment is a critical phase that requires careful planning and execution. The decommissioning of a wellbore is a tradeoff between cost, efficiency, and environmental responsibility. Poorly executed wellbore abandonment can become a burden to remediate and can result in regulatory and reputational damages for the operator or industry. This becomes even more critical if the abandoned wellbores are near communities or in sensitive areas. As such the industry and operators have focused on wellbore and decommissioning practices to ensure well abandonment procedures provide permanent barriers that prevent formation fluids from reaching the surface or cross flow between formations along the wellbore. Well barriers need to be able to withstand downhole conditions, stresses, and aggressive formation fluids, like CO₂ or brines.

Schmitz (1996) performed a study in Western Canada and concluded that an estimated 45% of the abandoned wellbores were experiencing gas to surface. The leak path could take place through the surface casing itself, the annulus, or the soil immediately surrounding the near wellbore.

Regulatory and Industry Guidelines Wellbore Abandonment

The decommissioning of oil and gas wells is a critical step in the lifecycle of a well, with the primary objective being the permanent isolation of all potential flow zones in the wellbore. API ST65-2 defines a potential flow zone as "any zone in a well where flow is possible when wellbore pressure is less than pore pressure". This definition applies to gas, oil, water, or fines migration. Proper wellbore abandonment ensures that hydrocarbons, water, or other fluids do not migrate to the surface or between formations, protecting both the environment and public safety.

In most countries, there are local rules and regulations that provide the requirements for well abandonment. The industry also has guidelines, such as API RP 65-3 or NORSOK D-10, that provide recommendations for best practices. Based on local regulatory requirements and industry best practices, operators and service companies have created internal standards and procedures on wellbore abandonment, well integrity, and verification of barriers.

Region field

The Region field oilfield is the largest onshore oil and gas field in the United Arab Emirates (U.A.E.). It has been discovered in 1958 and brought into production in the early 1960's. The field is located approximately 160 km southwest of Abu Dhabi, as shown in Fig. 1. The field spans an area of 1,200 km². Although the field is still in production with ongoing drilling activity, the field also contains numerous wells that are no longer productive and economic. Those wells are being abandoned as per local regulatory requirements and operator standards. A review of the abandonment procedures was initiated for improvements during the process.



Figure 1—Location of the Region Field is highlighted

Historical Decommissioning Process and Challenges in the region

Historically, wellbore abandonment in the region followed established practices that involved the placement of multiple abandonment cement plugs to isolate the reservoir. These included:

- A cement plug to isolate the reservoir
- A possible cement plug to isolate the top of the production liner
- A surface or environmental plug to provide additional isolation

In cases where there were additional shallow reservoirs, poor casing integrity, or inadequate cement integrity behind the casing, further steps were required. These included casing milling or casing retrieval, followed by the placement of additional barriers as needed. Fig. 2 illustrates a typical wellbore abandonment scenario in the Region field.

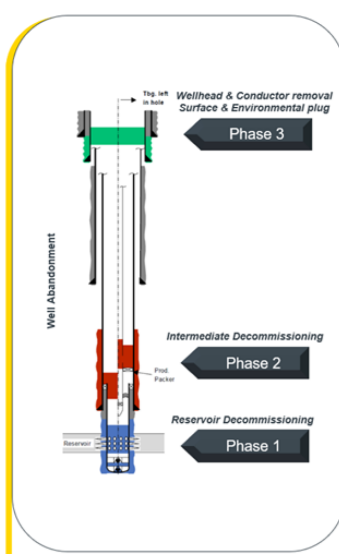


Figure 2—Typical wellbore abandonment used in the Region fields

Cement plugs are designed using Portland cement. Between the cement plugs, inhibited aqueous fluids or sometimes not treated fluids were left in the wellbore.

The review of the wellbore abandonment procedures was conducted proactively, as some of the wells being decommissioned are located in an area identified as strategic and sensitive, with potential social usage. Properly plugging and abandoning wells prevents the leakage of hydrocarbons and other contaminants into groundwater and surface. This helps protect ecosystems and drinking water sources at the same time while preventing unplanned emissions and flows. Also, the impact on the wellbore decommissioning regarding sustainability objectives is reviewed and consequently all the best practices and improvements are implemented.

Optimized Abandonment Procedure

The review of the wellbore abandonment and decommissioning procedure was initiated with the aim to improve the overall process and results, including wellsite efficiency, long-term abandonment goals, and aligning with sustainability targets. Below is a summary of the improvements and their impact.

- Reinforced and continue the implementation of the full well cement isolation and tubulars corrosion evaluation with identification of any required remediation actions due to corrosion or poor isolation behind casing.
- The strategy was evaluated and derisked completely; after a technical, operational, efficiency and cost analysis it was decided to eliminate any aqueous fluids in the wellbores scheduled for permanent decommissioning, so no liabilities are left behind due to ever-changing down hole conditions in the future and potential seismic or sub-surface events. Solid plugging material implementation is enforced across the full wellbore from bottom to top. Exceptional cases can have a dispensation if no accessibility is possible, in these cases the wellbore is recommended to be left with inhibited and corrosion treated fluids across those unique sections if possible.
- Custom designing the cement plug quality, effectiveness and cost by focusing on cement plug slurry qualifications and long-term mechanical properties.
- Improve cement plug mixing operations, recording, monitoring and placement techniques with QA/QC process and checks during the strictly batch mixing execution of the cement jobs, based on very detailed simulation and down hole conditions definitions in accordance with specialized software and global oil and gas best practices.

Well Integrity evaluation

The foundation of a robust and effective wellbore decommissioning procedure is a comprehensive evaluation of the actual wellbore downhole and surface conditions. Casing inspection logs will determine any casing deterioration during the active life of the well. Well integrity logs, examples of which are shown in [Fig. 3](#), will also confirm the presence of a barrier between the casing and the formation. Based on the results of the logs, the decommissioning procedures are adjusted to meet or exceed the requirements.

If the casing and isolation behind the casing are in good condition, a straightforward cement plug can be placed across the inside of the casing providing a continues reliable barrier. If the evaluation shows no barrier presence or if the barrier has failed over time, additional steps will have to be taken during the decommissioning. This can include squeezing cement to restore the barrier to section milling and placing a new barrier to restore rock-to-rock isolation.

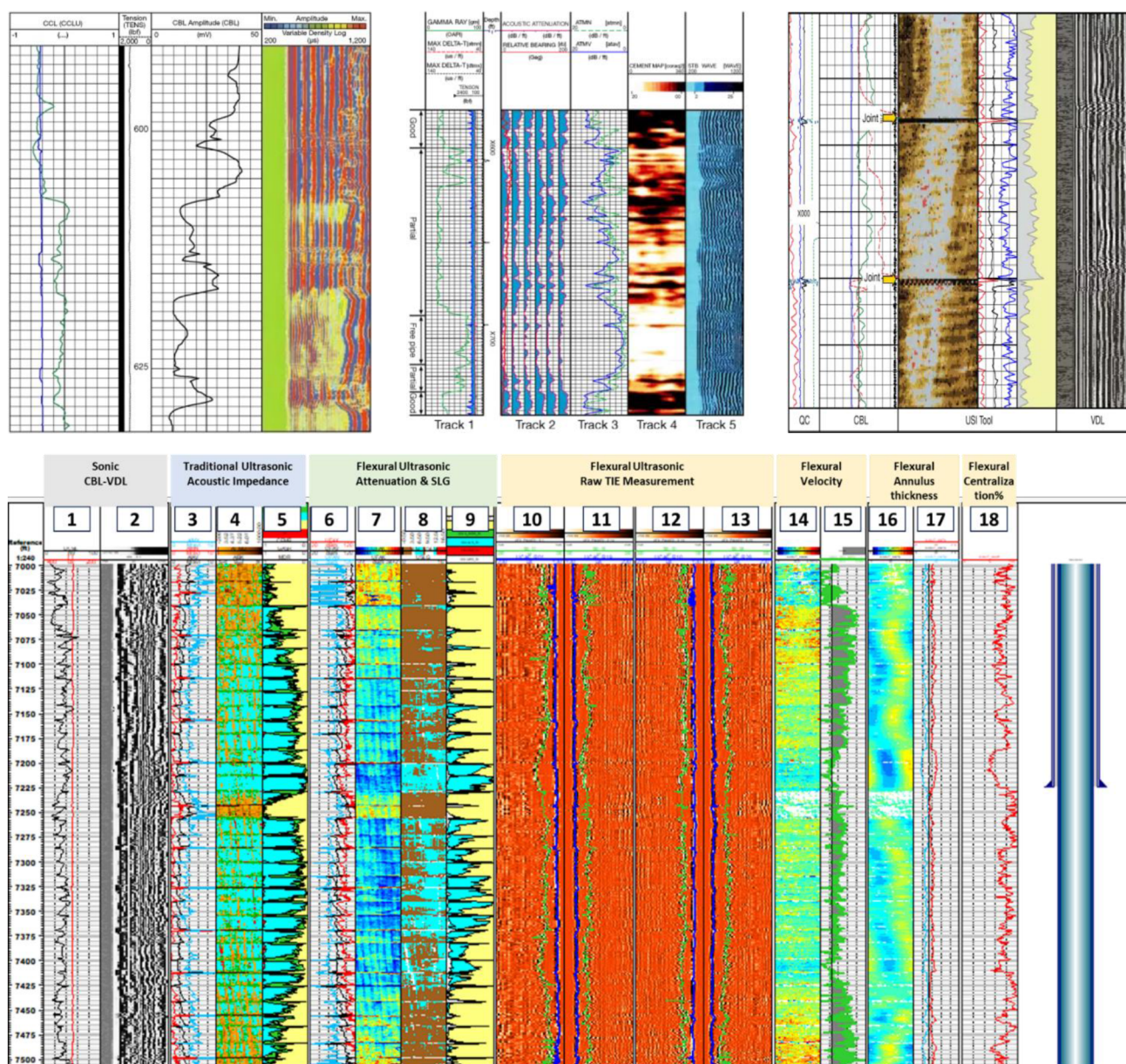


Figure 3—Evolution of the cement isolation evaluation in general, these technologies are routinely used across new wells and wells scheduled for decommissioning to ensure the zonal isolation between productive zones and required sensible zones as previous casing shoes, top of the liner, reservoirs, aquifers, cap rock. Accurate and reliable technology used for modern abandonments enables better decisions.

Wellbore Isolation

This detailed explanation highlights the challenges and considerations involved in ensuring the long-term integrity of a wellbore during its decommissioning process, with a particular focus on corrosion risks and chemical treatment. Leaving aqueous fluids in the wellbore can pose a risk, despite being professionally treated with oxygen scavenger and corrosion inhibitors with the correspondent chemicals using certified quality water on location.

Standard laboratory corrosion tests includes the salt spray test, (ASTM B117), which simulates brines from the aquifers exposure to the casings; immersion test (ASTM G31) for different casing materials in contact with liquids; electrochemical test for evaluating the corrosion resistance; stress corrosion test for cracking, pitting corrosion (ASTM G48); galvanic corrosion (ASTM G71) were done for the typical well conditions and cases in the internal and external laboratories to check on the efficiency and effectiveness

of the different commercially available corrosion and oxygen treatments under the typical laboratory and field conditions.

The laboratory results were diverse, ranging from minimal corrosion to immediate corrosion. However, the testing demonstrated that effective preservation is possible when the environment and process completely isolate external factors. This level of control, however, is challenging to achieve in field conditions. The chemicals tested were found to be a significant variable affecting the results, but the procedure for treating the fluids emerged as the primary factor influencing the variability of outcomes when simulating field conditions.

Metering, addition and the handling of the oxygen scavenger and corrosion inhibitors and the mix in the field is challenging due to the type of equipment available in the average drilling units, mixing tanks, centrifugal pumps, treating lines, connections, hoses, fluid transfers, pumping equipment. Many steps in the process have a potential risk where the aqueous fluid can be exposed to contamination or air entrapping, reducing the effectiveness of the chemical treatment.

To ensure the wellbore remains isolated in the future, a proactive approach has been adopted to fully encapsulate the casing with cement, both on the annular side and within the interior of the casing. Fig. 4 illustrates the evolution of cement plug placement in the new decommissioning procedure. By fully encapsulating the casing with well cement, the risk of casing corrosion is minimized to a negligible level. This approach significantly extends the lifespan of the barriers, ensuring the wellbore is responsibly plugged and abandoned in a state as close as practicable to its original condition before the well was drilled.

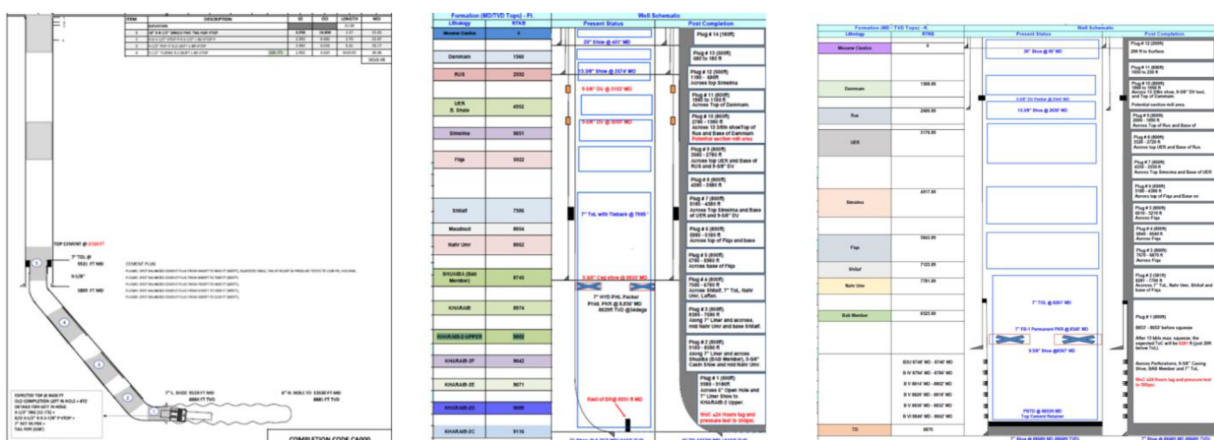


Figure 4—Evolution of P&A practices and an overview of the difference between recent years and current P&A strategy covering the full wellbore.

Cement plug fluid design

Previous abandonment cement plugs were designed with slurry properties matching the job requirements. Class G cement with silica at deeper section to overcome strength retrogression due to elevated temperatures, while Class G cement for abandonment plugs higher in the wellbore. The slurry was designed with fluid loss or gas-migration additives as needed when exposed to the reservoir. Dispersant and retarder were used for surface mixability and pumping requirements.

The required slurry properties were reviewed and optimized, with additional focus on the set cement mechanical properties. For the cement to withstand downhole stress and maintain isolation in the future, a new slurry design was introduced. The locally developed cement slurry features a lower Young's modulus, allowing it to accommodate stresses without compromising or losing integrity. An expanding agent was used to avoid cement shrinkage during the hydration process and to enhance bonding between the cement and casing. The cement design is extensively validated in the laboratory to meet all requirements. For the

latest solution, the silica flour blend is prepared and sourced locally and can be transported, stored, mixed and pumped with conventional cementing equipment.

Besides the cement blend, the laboratory testing and the job simulation helped to make the decision to implement the use of Class G with 35% BWOC (by weight of cement) silica flour for all plugs required to cover the wellbores from bottom to top with cementitious material as shown in Fig. 4. This avoids having many different blends on locations and eases operations and logistics. This blend is typically used only at elevated temperatures to overcome strength retrogression but can be designed as needed for lower temperatures. Besides the benefits of the rig site operations, the addition of silica flour also reduces the cement content in the cement slurry while providing higher stability, better long-term mechanical properties and durability.

Additional to this, as cement is a big contributor to greenhouse gases during its production, the addition of silica lowers the greenhouse gas emissions and provides benefits on the overall CO₂ footprint.

Wellbore Abandonment

In the new process, the entire wellbore is filled with cement, significantly increasing the number of cement plugs. To improve operational efficiency while maintaining a high standard of execution, a comprehensive review on the placement design and execution of cement plugs was conducted. To minimize the risks associated with setting cement plugs, the plug length is usually limited. Setting too small cement plugs can lead to excessive contamination of the cement and results in the cement not providing the isolation expected. Setting too long cement plugs increases the risk of stuck pipes. A balance must be found between setting too many plugs as this requires excessive rig-time and taking high risk. A detailed review of the placement procedures and slurry lab testing requirements provided an optimum length of cement plug, minimizing the risks. Using proprietary software, placement was optimized and cement contaminated minimized. During lab testing, slurry parameters were optimized to ensure safe placement and avoid the drill pipe getting stuck. Also, the placement technique was reviewed. Historically most cement plugs were placed as balanced cement plugs, but other placement techniques were considered such as sacrificial tubing and plug placement tools.

To address the increased number of cement plugs and avoid delays while waiting for them to set, a decision tree was developed to determine which cement plugs would be tagged and confirmed as well barriers. Other plugs were placed continuously but were not individually verified. Rig operations, particularly those involving cement plug placement, were reviewed and documented. Post-job documentation and analysis ensured that on-site execution adhered to the agreed-upon plan. If discrepancies were noted, corrections were made to the procedure to prevent similar issues in future operations.

During the P&A operations it was observed that typically the cement slurries were batch mixed using a pressurized mud balance, given the volumes required for each plug. The improvements allow for the cement slurry to be double checked, verified, confirmed while recorded and monitored using the typical cement density gauges and instrumentation available on location since it is very important and understood the control and delivery of precise density mixing and measurement in the slurry preparation as illustrated in Fig. 5, which is critical for maintaining the long term performance and safety of the well decommissioning operations. Emphasis was placed on double-checking the mass balance during slurry design to ensure accurate proportions of water, additives, and cement. It also was stressed the importance of gradual slurry weighting and frequent density checks under pressure. The key takeaway is that any variation in density can impact well control, fluid quality, and mechanical properties, making continuous monitoring essential to deliver the expected fluid performance and maintain operational safety, integrity, consistency, accuracy and quality. With this process the cement systems are recirculated in the mixing equipment until the optimum specifications are achieved and confirmed ready for deployment.

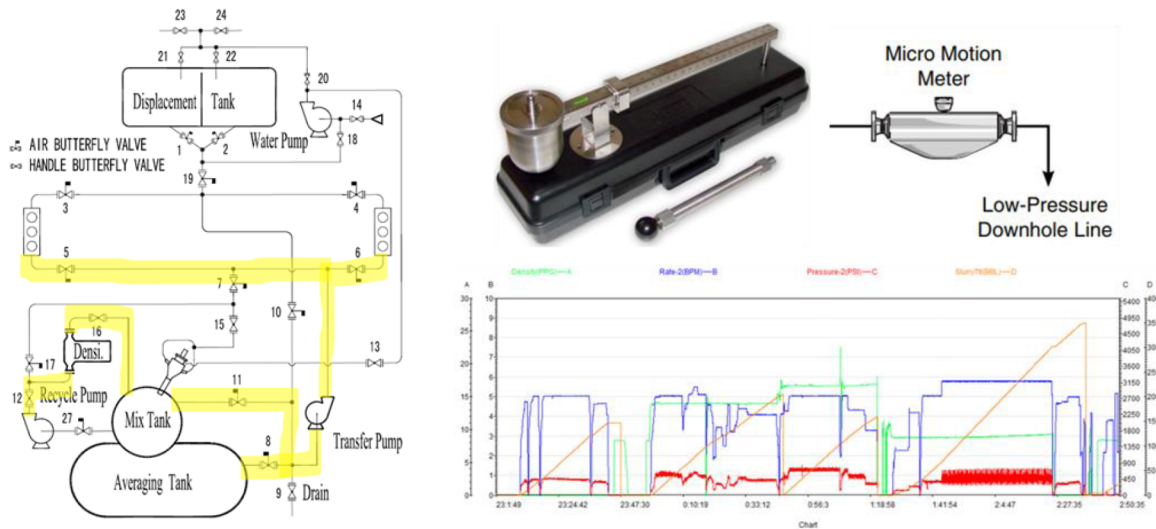


Figure 5—Equipment layout and set up adopted during batch mixing operations, the mud balance and the online fluids density meter along with the typical chart recording and monitoring the density of all fluids before being placed on the wellbore.

Validation and Regulatory Compliance

To ensure validation and regulatory compliance for plug and abandonment (P&A) barriers, several key documents and communications provide detailed guidelines and procedures.

Most of the key documents not only in the region but globally comprise the internal or operator's guidelines for well integrity management standards and manuals, this document outlines the Well Integrity Management Standard (WIMS), which includes comprehensive guidelines on well barrier elements, their validation, and regulatory compliance. Key sections typically include:

- Common well barrier elements
- Annular cement well barrier elements
- Well barrier design, selection, and construction
- Validation of well barrier elements
- Pressure test direction
- Leak testing of well barrier elements
- Target pressure values
- Acceptable leak rates and test duration
- Documentation of leak and function testing of well barriers
- Well barrier monitoring
- Well barrier elements
- Acceptance criteria (EAC) tables

Another document frequently used by the technical and operational teams is the wellbore annular cement evaluation manuals or guidelines. This document includes best practices for evaluating wellbore annular cement, which is crucial for ensuring the integrity of P&A barriers reviewing:

- Operator Codes of Practice

- Cement Bond Logging Techniques and Interpretation
- Guidelines for Cement and Zonal Isolation Acceptance and Approval
- Factors Affecting Cement Bond Log Quality and Interpretation
- Remedial Action Cement Squeezes

Cementing Slurry Considerations

The process of designing cement slurry for plug and abandonment (P&A) operations is critical to ensuring well integrity, long-term durability, and environmental protection. The fluids and plugging materials should be designed to provide sealing of any fluid from the reservoir or avoid any contamination to the freshwater banks and eliminate any possible cross flows between different pressures intervals. The systems should have a good bonding specification to the tubulars and to formation accordingly. And they should be able to endure bottom hole conditions and have long durability since the wells are decommissioned for the rest of their life. These conditions can generally be met using a 15.8 ppg cement system as API Class G with the required chemicals according to the bottom hole conditions.

Preferred cement systems are generally accepted because they are less susceptible to mud contamination if proper cementing practices are followed, minimizing strength loss due to contamination. These systems also set and gain compressive strength more quickly than other alternatives, thereby reducing the waiting-on-cement (WOC) time. Additional cement additives must be optimized to suit the specific application, addressing critical properties such as fluid loss, rheology, pumpable time, hydration time, compressive strength, and dehydration of the cement.

In general, but depending on the final individual geomechanics, geographical zones, the areas with non-corrosive water, moderate sulphate resistant cement should be recommended, API Class G or H. If the zones are prone to high sulphate or acid waters presence, a special high- sulphate resistant cement system should be available to provide higher resistance to cement leaching.

Selecting or designing a cement system is a function of the final well conditions, depth, temperature, wellbore fluids, surface equipment and pumping strategy. Special conditions in the well such as losses, salt formation, shales, etc should be considered when preparing the cement system. Defining well conditions, selecting an appropriate cement type, adding water, specialized additives or components and conducting laboratory tests to achieve desired slurry properties like density, rheology, fluid loss, thickening time, and compressive strength. This iterative process ensures the cement slurry can effectively seal and support the wellbore casing, isolate different geological zones, and very important provide long-term or for life well integrity under various downhole temperatures and pressures.

Typically, the preferred slurry selection due to stability, mechanical properties, environmental impact, economical and long term endurance, is the silica flour blend-based slurries which is routinely used at the moment. It can be designed and adjusted for the different downhole temperatures, customized to the different job times along with the thickening requirements.

Environmental Considerations

It is essential to take all precautions while planning to design a responsible P&A strategy. Ensuring that the well is properly plugged with cement achieving zonal isolation to eliminate any possible interzonal cross flows. The plugging methods and cementing operations carried during P&A activities will have a direct impact on the environmental conformance and many concerns can be observed if they are not performed satisfactorily and as per local and industry recommendations. The failure to completely seal across the wellbores would leave a potential for fluids migration across different pressure gradients zones.

Globally, several cases have been documented where inadequate or improper isolation during P&A operations resulted in undesired and unplanned formation fluid flows through the wellbore and to the

surface, releasing greenhouse gases that contribute to global warming. Many of these cases are associated with older assets, some of which were abandoned over a century ago using techniques that are no longer considered optimal by modern oil and gas standards. Additionally, studies have identified unexpected and sudden increases in reservoir pressure caused by recent secondary recovery operations—factors that were not accounted for during the original wellbore abandonment process.

While the history of P&A operations demonstrates significant improvements and technological advancements in recent years, there remains considerable room for improvement. Incorporating all potential scenarios, including worst-case events such as natural disasters, into the planning, execution, and monitoring of permanently abandoned or decommissioned wells is critical to ensuring long-term well integrity and environmental protection.

Additional tools, materials and technologies

Very recent new technologies are targeting better well decommissioning to ensure the rock-to-rock isolation of the wellbores across the cap rocks, so the original reservoir fluids migration is eliminated. These technologies are still under piloting and have not been widely adopted as practices across the industry.

Case Study – Well A

The well presented here is an example and guidance for responsible mature asset decommissioning shown in Fig.-6. The well was drilled to 9210 ft in 1980, with a 20in, 13 3/8in, 9 5/8in casings and a 7in liner and completed with 2 7/8 in × 3 1/2 in tubing and packer. The P&A highlights called for moving the rig to location, check for pressure, kill the well, recover old completion, record corrosion and cement evaluation for the entire well, plug back and abandon the well, hand over for monitoring the well at surface for pressure across the annular spaces and the main well bore.

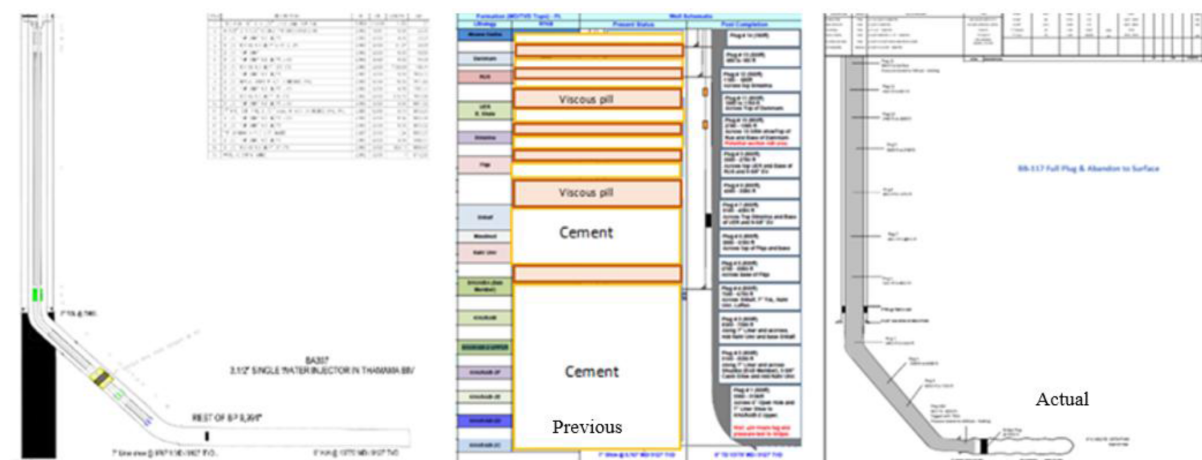


Figure 6—Original down hole well status, typical well P&A former outdated P&A proposal in orange showing cement next to the well formations, viscous pills on brown. The latest practice with single cement plug from TD to surface update procedure with final well DHWS.

Traditionally the cement plugs for abandonment were only placed in a minimum required length and depth. While this could show an extremely efficient approach, it has been demonstrated that this approach has the potential for high risk of unplanned events developing in the future. Due to the downhole conditions and time decreasing the integrity of the well barriers due to several reasons such as seismic events, corrosion, naturally occurring reactions, field development strategies allowing for migration paths for particularly gas or even fluids.

The more reasonable and responsible method in place is to inspect the asset and the integrity elements on surface and downhole, run the required tools for evaluation of the barrier and the full envelop to clearly

understand the current situation of the asset, review the mechanical status and define any limitations or potential risks. Based on this initial analysis, the final strategy and operational procedures are designed to align with the strict adopted methodology. Any remedial treatment is performed, tested and qualified before the final abandonment is started and follow by accessing the lowest most accessible depth of the well to set the first plug and continue to create a single and uninterrupted cement barrier along the full length of the well, this provides extremely higher possibilities to permanently isolate and eliminate any unexpected potential gas or fluids migration.

Improving cement plug practices for modern decommissioning and including controls such as tagging cement and pressure testing after a minimum of 24 hrs (WOC) without any disturbance from DHT is a key parameter to evaluate the effectiveness of the plug and abandoned program. The results are very clear when they are measured so the control and monitoring of the decommissioned assets with pressure sensors at surface, well head inspections and leak detection for several years successfully can eliminate or significantly reduce the exposure to undesired pollution or gas/fluids migration to surface.

Conclusions

Responsible and safe plugging and abandonment of oil and gas wells require a meticulous well integrity assessment, installation of permanent barriers, such as cement aided by mechanical down hole tools as required or necessary to prevent leaks, thorough verification of barrier effectiveness, and documentation for long term risk reduction. Key challenges include dealing with incomplete data on older wells, managing environmental risks like groundwater contamination or seabed pollution, and the increasing urgency due to the large number of wells approaching end-of-life. Successful P&A hinges on skilled technical engineering, proactive planning to avoid costly deferred work, and a commitment to environmental stewardship beyond mere compliance.

This new wellbore decommissioning procedure has now been standardized in the Region and nearby fields. Non-economic wells are safely and responsibly plugged and abandoned, leaving no concerns regarding the long-term environmental impact of retired assets, particularly to those close to social centers and with any risk of hydrocarbon emissions. The novel and optimized approaches provided both a time saving of 36% and by optimizing the fluid requirements and logistics, the total well abandonment costs was reduced by 28%. The onshore operator team recognized the methods as cost-effective to retired this well and future assets.

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